Design and Verification of a Single-Cycle RV32I Processor in Verilog

Phase I Report — CSA-222 Modern Computer Architecture

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***Abstract*—This report describes Phase I of a project to build a compact single-cycle processor that executes a ten-instruction subset of the RISC-V RV32I base instruction set: ADD, SUB, AND, OR, XOR, ADDI, LW, SW, BEQ and BNE. The processor is written in synthesizable Verilog as eleven small modules that mirror the textbook datapath: a program counter, instruction memory, control unit, register file, immediate generator, ALU with its own decoder, data memory and the adders and multiplexers that connect them. Every module has a self-checking testbench, and the integrated processor has been simulated in Icarus Verilog running two hand-assembled test programs. Both programs finish with the register and memory contents that were worked out by hand beforehand. Phase II will widen the test coverage, synthesize the design and report resource usage and timing.**

***Index Terms*—RISC-V, RV32I, single-cycle processor, datapath, control unit, Verilog**

I. Introduction

Every program eventually runs as a sequence of fetch, decode and execute steps on a processor. The simplest hardware that can do this is the single-cycle processor, in which each instruction completes within one clock period. It is not how fast commercial processors are built, but every later idea in the course, including pipelining, hazards and forwarding, is a modification of this design, so building it is the most direct way to understand the others.

We chose RISC-V as the instruction set because it is an open standard, its base integer subset is small, and its instruction formats were designed to make decoding simple. The course labs introduce datapaths through MIPS32; RISC-V follows the same load-store philosophy, so the concepts carry over directly while the encoding details are new.

The goal of the full project is a working processor with RTL, testbenches, waveforms, test programs and a synthesis report. Phase I covers the design, the RTL for every module and simulation of the integrated processor.

II. Literature Review

A. RISC and the RISC-V instruction set

Patterson and Ditzel argued in 1980 that a small set of simple, regular instructions leads to simpler hardware and a shorter design time than a large complex instruction set [1]. RISC-V, started at UC Berkeley in 2010, applies that idea as a free and open standard [2]. The specification defines a small base integer ISA, RV32I, and keeps everything else (multiplication, floating point, compressed instructions) in optional extensions [3]. RV32I has 32 registers of 32 bits with `x0` fixed at zero, fixed 32-bit instructions, and a handful of formats in which the register fields and the sign bit of the immediate always sit in the same bit positions. These choices matter for this project: the register file can be read before the instruction type is known, and sign extension always uses bit 31.

B. Textbook single-cycle designs

Patterson and Hennessy develop a single-cycle datapath step by step for a subset of load, store, arithmetic-logical and branch instructions, and introduce the two-level control scheme in which a main decoder produces a 2-bit ALU operation class and a second small decoder turns it into the exact ALU function [4]. Harris and Harris present a similar RISC-V single-cycle processor and give complete HDL for it [5]. Both texts point out the main weakness of the approach: the clock period has to fit the slowest instruction, the load, so every other instruction wastes time. This is the motivation for multi-cycle and pipelined designs, which are outside the scope of this project. Nisan and Schocken take the same build-it-from-small-parts approach from logic gates up to a complete computer [6], which influenced how we split the work into small independently testable modules.

C. Open-source RISC-V cores

Several open cores show what the design space looks like beyond the textbook. PicoRV32 is a size-optimized multi-cycle core that trades cycles per instruction for a small area and a high clock frequency [7]. SERV goes further and processes data one bit at a time to minimize area [8]. Ibex is a production-quality two-stage pipelined core with extensive verification [9]. These cores optimize for area, frequency or robustness and are correspondingly harder to read. Our design optimizes for clarity instead: one module per block of the datapath diagram, no pipelining, and only the instructions needed to demonstrate arithmetic, memory access and control flow.

D. Tools

The design is simulated with Icarus Verilog [10] and waveforms are viewed in GTKWave [11], the same tool flow used in the course labs.

III. Proposed Solution

A. Instruction subset

Table I lists the ten supported instructions. They cover four of the RV32I formats. R-type instructions operate on two registers. I-type instructions (ADDI and LW) take one register and a 12-bit immediate. The S-type store splits its immediate across two fields so that the register fields stay in place. The B-type branches encode a 13-bit offset whose lowest bit is always zero.

TABLE I
Supported instructions and their encodings

Instr.	Format	opcode	funct3	funct7
ADD	R	0110011	000	0000000
SUB	R	0110011	000	0100000
AND	R	0110011	111	0000000
OR	R	0110011	110	0000000
XOR	R	0110011	100	0000000
ADDI	I	0010011	000	—
LW	I	0000011	010	—
SW	S	0100011	010	—
BEQ	B	1100011	000	—
BNE	B	1100011	001	—

B. Datapath

Fig. 1 shows the datapath. One instruction moves through it from left to right in a single clock cycle.

Fetch. The program counter (PC) holds the address of the current instruction. The instruction memory is read combinationally, and an adder computes PC + 4.

Decode. The opcode goes to the control unit. The `rs1` and `rs2` fields address the register file, which returns both values combinationally. The immediate generator extracts and sign-extends the immediate according to the instruction format.

Execute. A multiplexer selects either the second register value or the immediate as the second ALU operand. The ALU adds, subtracts or performs AND, OR or XOR, and raises a `zero` flag when the result is zero. A second adder computes the branch target PC + immediate.

Memory. For LW and SW the ALU result is the memory address. Data memory is read combinationally and written on the clock edge.

Write-back. A multiplexer selects the ALU result or the memory data and sends it back to the register file, which is written on the clock edge.

Only three things change on the clock edge: the PC, one register and one memory word. Everything else is combinational logic, which is what makes the design single-cycle.

C. Control

Control is split into two levels, following [4]. The main control unit looks only at the opcode and produces the signals in Table II. The ALU control block then uses the 2-bit `alu_op` together with `funct3` and bit 5 of `funct7` to pick the ALU operation. ADD and SUB differ only in that `funct7` bit; for ADDI the same bit position belongs to the immediate, so subtraction is selected only when the opcode is R-type.

Branches reuse the ALU: it subtracts the two registers, and `zero` = 1 means they are equal. BEQ and BNE differ only in `funct3[0]`, so the next-PC select is

$$\texttt{pc_src} = \texttt{branch} \cdot (\texttt{zero} \oplus \texttt{funct3[0]}).$$

An unknown opcode leaves every write enable at zero, so it cannot corrupt registers or memory.

TABLE II
Main control unit truth table (x = don't care)

Class	reg_write	alu_src	mem_write	mem_to_reg	branch	alu_op
R-type	1	0	0	0	0	10
ADDI	1	1	0	0	0	10
LW	1	1	0	1	0	00
SW	0	1	1	x	0	00
BEQ / BNE	0	0	0	x	1	01

D. RTL structure

Each block in Fig. 1 is one Verilog module: `pc`, `adder`, `instr_mem`, `control_unit`, `reg_file`, `imm_gen`, `alu_control`, `alu`, `mux2`, `data_mem`, and the top level `rv32i_top` which only wires them together. Both memories hold 64 words. The instruction memory is initialized from a hex file with `$readmemh`, so a new program can be run without touching the RTL. Programs are written in RISC-V assembly and assembled by hand into hex; each hex line carries the assembly as a comment.

E. Verification and current status

Every module has a self-checking testbench that applies inputs, compares the outputs against expected values and prints a pass or fail summary. Table III summarizes them. All checks pass in Icarus Verilog.

The integrated processor runs two programs. Program 1 uses ADDI, ADD, SUB, AND, OR, XOR, SW and LW on the values 10 and 6, including a negative immediate, and reaches its halt instruction after 14 cycles. Program 2 adds the numbers 5 down to 1 in a loop and reaches halt after 24 cycles; it exercises BNE and BEQ in both the taken and not-taken directions and ends with a self-check that sets `x6` to 1 only if the sum read back from memory equals 15. Table IV compares the simulated final state with the values computed by hand. They match in every case.

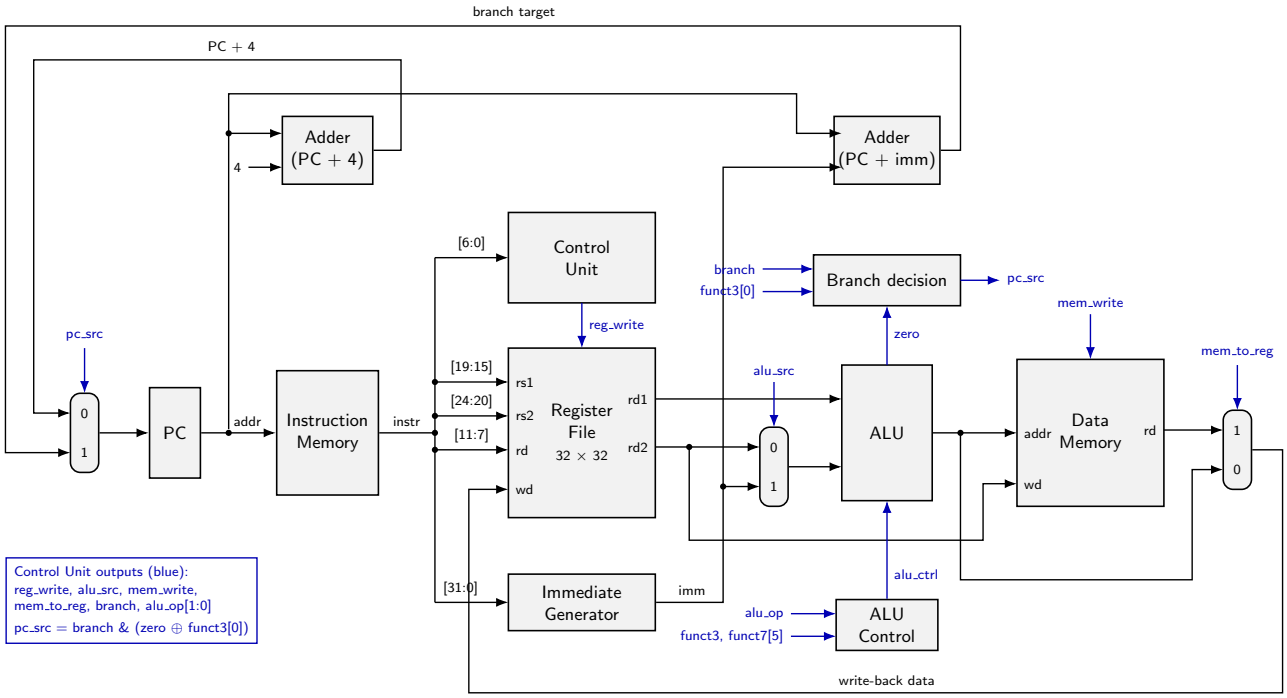


Fig. 1. Datapath of the single-cycle RV32I processor. Data signals are drawn in black and control signals in blue.

TABLE III
TESTBENCHES AND PHASE I SIMULATION STATUS

Testbench	What is checked	Checks
`tb_fetch`	reset, PC+4 stepping, jump to target, instruction read	7
`tb_reg_file`	write and read, dual read ports, write enable, `x0` stays zero	35
`tb_imm_gen`	I, S and B immediates, positive and negative	11
`tb_alu`	five ALU operations, zero flag, ALU decoder for all ten instructions	21
`tb_control_unit`	control table, unknown opcodes, multiplexer	9
`tb_data_mem`	write and read back, write enable, address range	9
`tb_rv32i_top`	program 1 and program 2 final state	23

TABLE IV
EXPECTED AND SIMULATED FINAL STATE

Program	Location	Expected	Simulated
1	`x3` (10 + 6)	16	16
1	`x4` (10 − 6)	4	4
1	`x5` (10 AND 6)	2	2
1	`x6` (10 OR 6)	14	14
1	`x7` (10 XOR 6)	12	12
1	`x8` (ADDI −3)	−3	−3
1	`x11` (sum of loads)	20	20
1	`mem[8]`	20	20
2	`x2` (loop sum)	15	15
2	`x6` (self-check)	1	1
2	`mem[0]`	15	15

F. Plan for Phase II

The remaining work is: (1) more test programs aimed at corner cases such as maximum and minimum immediates, forward and backward branches at the edges of the offset range, and writes to `x0`; (2) synthesis of the design and a resource report; (3) an estimate of the critical path and maximum clock frequency, with a discussion of why the load instruction sets it; (4) deployment on an FPGA board if one is available; and (5) the results and discussion sections of the final report.

IV. INDIVIDUAL CONTRIBUTIONS

The work was divided so that each member owns a part of the datapath together with its testbench, one section of this report and one segment of the demo video. Table V lists the split.

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[3] A. Waterman and K. Asanović, Eds., *The RISC-V Instruction Set Manual, Volume I: Unprivileged ISA*, Document Version 20191213, RISC-V Foundation, Dec. 2019.

[4] D. A. Patterson and J. L. Hennessy, *Computer Organization and Design RISC-V Edition: The Hardware Software Interface*, 2nd ed. Cambridge, MA, USA: Morgan Kaufmann, 2020.

[5] S. L. Harris and D. M. Harris, *Digital Design and Computer Architecture: RISC-V Edition*. Cambridge, MA, USA: Morgan Kaufmann, 2021.

TABLE V
INDIVIDUAL CONTRIBUTIONS IN PHASE I

Member	RTL and verification	Report and other
Member 1	`pc`, `adder`, `instr_mem`, `tb_fetch`; both test programs, hand-assembled	Abstract, Introduction
Member 2	`reg_file`, `imm_gen` and their two testbenches	Literature Review
Member 3	`alu`, `alu_control`, `tb_alu`	Instruction subset, Datapath, block diagram
Member 4	`control_unit`, `mux2`, `tb_control_unit`	Control section and truth table
Member 5	`data_mem`, `rv32i_top` integration, `tb_data_mem`, `tb_rv32i_top`	Verification section, repository, video editing

[6] N. Nisan and S. Schocken, *The Elements of Computing Systems: Building a Modern Computer from First Principles*, 2nd ed. Cambridge, MA, USA: MIT Press, 2021.

[7] C. Wolf, "PicoRV32: A size-optimized RISC-V CPU." [Online]. Available: https://github.com/YosysHQ/picorv32

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